

PARTICULAR SPECIFICATIONS P4E

Appendix 4E – Low Pressure Heating Water (LPHW) Supply Criteria

CONTENTS

P4E.1	Low Pressure Heating Water (LPHW) Supply Criteria.....	1
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P4E.1 LOW PRESSURE HEATING WATER (LPHW) SUPPLY CRITERIA

- a) The Hot Water Return (HWR) line from first stage cooling heat exchangers shall be transferred to Greasy Waste Receiving Facility (GWRF) for use as Low-Pressure Hot Water (LPHW). Table P4E-1 below shows the parameters to be achieved for supply of Hot Water to GWRF.
- b) As shown in Figure P4E-1 below, the hot water from Thermal Hydrolysis Plant (THP) will be transferred to GWRF's LPHW tank via the HWR supply line. The hot water will pass through the outer section of double pipe. This functions to keep inner pipe's skin temperature above ambient temperature via counter-current flow. Greasy Waste will flow within the inner pipe and the elevated temperature of the inner pipe skin will assist with the fluid flow. Before the greasy waste pipe, there is a tee-off to meet the intermittent hot water spray requirement for Greasy Waste Holding Tanks.

Table P4E-1: LPHW Design Basis

Parameter	Unit	Remarks	Value
HWR supply temperature at interface (TP-0101)	°C	-	50
HWR supply pressure at interface (TP-0101)	barg	Note 1	Min 4 barg at design flowrate
HWR supply flowrate at interface (TP-0101)	L/s	Note 2, 3	During GWRF operating hours: 29 L/s (105m ³ /h) - Continuous During GWRF off hours: Normally no flow 18L/s intermittent demand for Greasy Waste Holding Tanks hot water spray.
LPHW Arrival Temperature at GW (required)	°C	Note 4	40-45
Estimated draw rate from LPHW tank and ring main at Greasy Waste	L/s	Note 4	20L/s (Average) 40L/s (Instantaneous)
Greasy Waste Transfer Pipe Skin Temperature	°C	Note 4	40-45

Note

1. In order to achieve this, maximum allowable pressure drop across Heat Exchanger and control valves shall be <1bar.
2. The Contractor shall evaluate the rated flow based on the selected instrumentation and controls. Extra flowrate is envisaged to compensate the instantaneous fluctuation due to Temperature control of the HWR at each heat exchanger unit. A dump valve at the common supply line shall function to trim the excess flow (which shall return to TCD wet well).
3. GWRF operational hours are Monday to Friday, 08:00 to 17:00; Saturday 08:00 to 12:00. HWR shall be supplied during the operational hours of GWRF and additional 1 to 2 hours for pre-start and afterhours cleaning. During off-hours/days, the dump valve shall function to divert all flows to TCD wet well. Occasionally, it shall react to intermittent draws from Greasy Waste Holding Tanks spray (near digesters) via pressure monitoring on the supply line.
4. For information only.

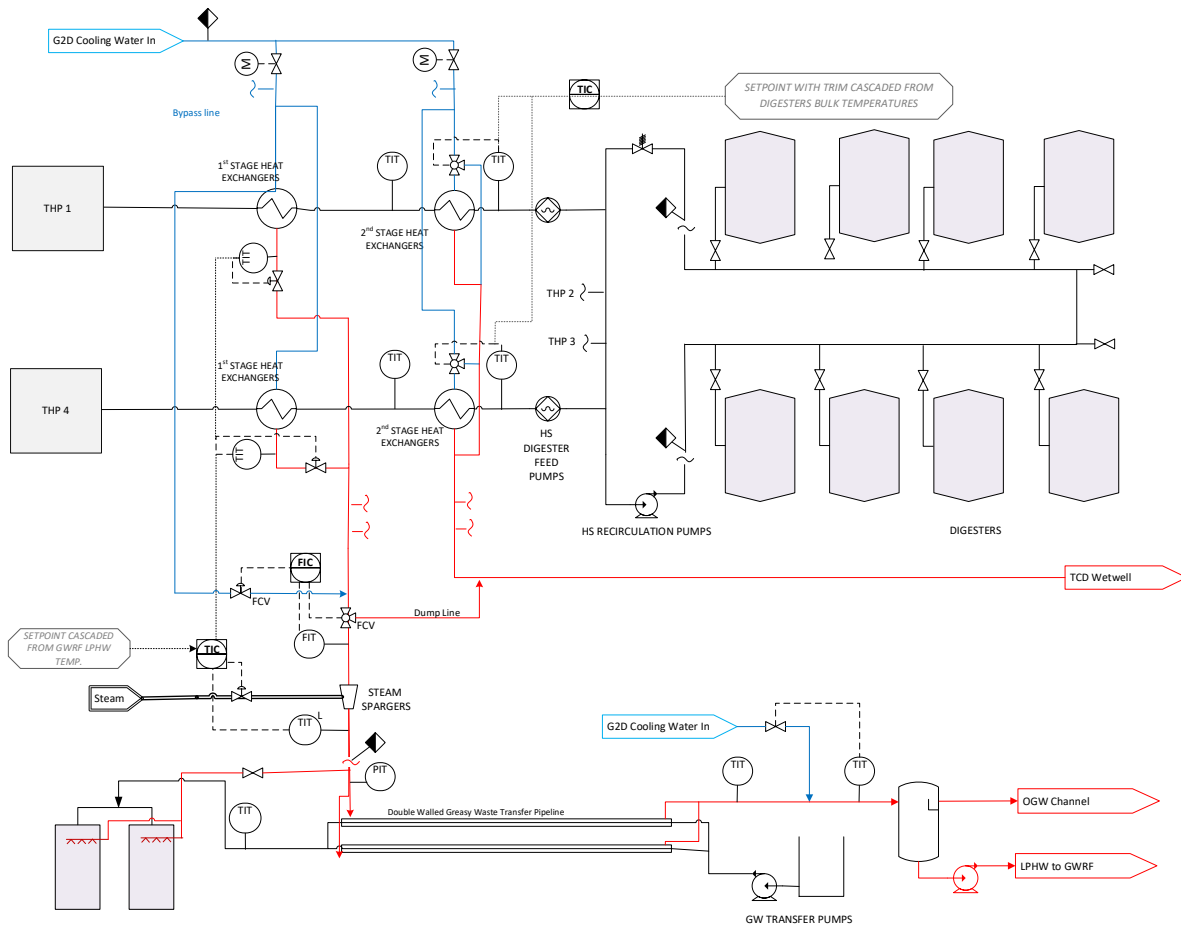


Figure P4E-1: Flow Schematic of Cooling Water and Hot Water Return for Reuse as LPHW

- c) Figure above shows the overall schematic for LPHW supply. As described in other sections of the specifications, there shall be a supplemental heating system (inline steam sparger) with dedicated bypass flow control to make-up the hot water supply demand in case of turndown of THP system and heat exchangers.

END OF SECTION